INCH-POUND
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PERFORMANCE SPECIFICATION

TANK AND PUMP UNIT, LIQUID DISPENSING, PORTABLE, PMU-29/E

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of dolly mounted, 3-gallon capacity, jet oil servicing tank and pump unit with 2-gallon overflow tank, designated PMU-29/E.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks of the exact revision listed below form a part of this specification to the extent specified herein.

FEDERAL STANDARDS

FED-STD-595/24052 Green, Semigloss

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-7808	Lubrication Oil, Aircraft Turbine Engine, Synthetic Base
MIL-PRF-8188	Corrosion-Preventive Oil, Gas Turbine Engine, Aircraft Synthetic Base
MIL-DTL-8795	Hose Assemblies, Rubber, Hydraulic, Fuel and Oil Resistant
MIL-PRF-23699	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Numbers: O-152, O-154, O-156, and O-167
MIL-DTL-25677/1	Adapter, Pressure Lubricating Oil Servicing, Aircraft
MIL-DTL-25677/3	Nozzle, Pressure Lubricating Oil Servicing, Locking

(Copies of these documents are available online at http://quicksearch.dla.mil/ or https://assist.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. This specification makes provisions for first article testing (see 4.4).

3.2 Materials. The contractor shall select the materials. Cadmium (including plating) and magnesium (including alloys) shall not be used. All materials shall be compatible with lubricants conforming to MIL-PRF-7808, MIL-PRF-8188 and MIL-PRF-23699 and shall be capable of meeting all of the operational and environmental requirements specified herein. Dissimilar metals shall not be used in intimate contact with each other unless protected against electrolytic corrosion.

3.2.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum

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extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Design and Construction. The tank and pump unit shall dispense oil to aircraft lubricating oil systems requiring pressure servicing with synthetic base lubricating oils. The unit shall permit convenient, one-man operation. The unit shall maintain oil initial cleanliness with respect to particulates and water under all operating environments and circumstances. Unit exterior shall have no feature that might trap fluids. Parts subject to loss require a means of being retained on the unit. The unit shall have a 2-gallon overflow capacity. The unit shall be dolly mounted. Unit parts shall not work loose in service and shall withstand the stresses, jars, vibrations and other conditions incident to shipping, storage, movement and operation.

3.4 Pump.

a. The pump shall be self-priming.

b. The pump handle shall be conveniently accessible and located so that the operator has sufficient operating room during pumping. The pump handle may serve as a means of carrying the supply tank when it is removed from the dolly or another means of carrying the supply tank shall be provided.

c. A means to ensure positive pumping of oil so that no air is pumped shall be provided.

d. The pump shall meet the pumping requirements specified in 3.11. An adjustable leverage principle may be incorporated to meet the pumping requirements under the varied temperature conditions.

e. A means of relieving pressure in the discharge hose shall be provided in the pump assembly so that the filter and discharge line remain flooded with oil between servicing, but pressure is relieved.

3.5 Filter. A micron type hydraulic filter assembly and filter element meeting the flow and pressure requirements of 3.11 shall be installed between the outlet of the pump and the inlet of the discharge hose. The filter assembly shall not include a by-pass valve.

3.6 Hoses and Nozzles.

a. The hoses shall be a minimum of 6-foot long and shall conform to MIL- DTL -8795. The MIL-DTL-8795 size 10 dispensing hose shall have a MIL-DTL-25677/3 nozzle at the discharge end of the hose. The MIL- DTL -8795 size 12 return hose shall have a MIL-DTL-25677/3 nozzle for attachment to the engine overflow adapter of the aircraft

b. A means to keep the nozzles clean when not in use shall be provided.

c. A means of securing the hoses on the unit for storage shall be provided.

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3.7 Dolly. The dolly furnished as an integral part of the unit shall have a minimum of three (3) wheels, a stabilizing brake and a tow bar suitable for manual towing or attachment to a tow vehicle.

3.7.1 Interface. Tow bar shall be compatible to tow vehicles with a pintle height between 10 inches and 30 inches high.

3.8 Adaptability. A means of adapting the unit for the servicing of aircraft with gravity-fill oil tanks shall be provided.

3.9 Component Removal. All major components shall be easily removed, replaced, and operated under all operating environments and circumstances with the use of common hand tools.

3.10 Interchangeability. All major component parts shall be interchangeable on all like models.

3.11 Performance. At 70º ±5º F ambient temperature and with its MIL-DTL-25677/3 delivery nozzle connected to a MIL-DTL-25677/1 adapter, the unit shall dispense oil conforming to MIL-PRF-7808 at a rate of not less than 1 gallon per minute with not more than 50 strokes or revolutions per minute of the pump handle. At not more than -40 ºF and with its MIL-DTL-25677/3 delivery nozzle connected to a MIL-DTL-25677/1 adapter, the unit shall dispense MIL-PRF-7808 oil at a rate of not less than 0.25 gallons per minute with not more than 50 strokes or revolutions of the pump handle and with not more than 50 pounds exerted at the pump handle grip. At not more than -40ºF and with not more than 50 pounds exerted at the pump handle, the unit shall develop not less than 300 pounds per square inch (psi) deadhead pressure of MIL-PRF-7808 oil. The unit shall operate under all conditions and environments without leaking.

3.12 Environmental conditions. The tank and pump unit shall operate and maintain oil initial cleanliness under the following conditions:

a. Temperatures ranging from -80º F to +160º F (See para. 4.4.7.1 and 4.4.7.2).

b. Relative humidity of 95% to 100 % at operating temperature of 125ºF and storage temperature of 160ºF (See para. 4.4.7.3).

c. Sand and dust particles as encountered in normal and desert operations (See para. 4.4.7.5).

d. Rainfall as encountered in any locale (See para 4.4.7.4).

3.13 Storage temperatures. The tank and pump unit shall withstand storage temperatures ranging from -80ºF to +160º F.

3.14 Pressures. All components downstream of the pump assembly, including the fill nozzle, shall withstand a test pressure of 500 psi or that deadhead pressure produced with a force of 150 pounds at the pump handle, whichever is greater. The supply and overflow tanks shall withstand an internal pressure of 10 psi or 200 percent of the operating pressure, whichever is greater. A

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means to relieve the pneumatic operating pressure and to depressurize the supply tank after operation shall be provided.

3.15 Mobility. The tank and pump unit complete with all components and mounted on its dolly shall meet the following requirements:

a. Lift points shall be located so as to ensure the unit is 2-man portable when empty.

b. With a full supply tank, the unit shall be capable of being towed by 1-man.

c. With a full supply tank and using a towing vehicle with a 10 foot turning radius, the unit shall be capable of being towed without tipping at a maximum speed of 7.5 mph, and negotiating right and left circle turns at maximum cramping of the tow vehicle.

d. With a full supply tank, the unit shall be capable of being towed at a maximum speed of 7.5 mph, over 60 º angle obstructions which are 0.75 to 1-inch high without damage to the structure or components.

e. With a full supply tank, the unit shall withstand being dropped from a height of 1 foot onto a concrete floor without damage to the structure or components.

f. The unit mounted on its dolly shall have a low center of gravity to preclude tipping from natural upright position.

3.16 Reliability. The tank and pump unit shall have a minimum Mean Time Between Failures (MTFB) of 27,000 pump cycles.

3.17 Interface details:

a. Supply tank and overflow tank shall each have a clean out access which is a minimum of 6-inches in diameter.

b. Filler opening shall have a minimum inside clearance diameter of 1.50 inches.

c. An MIL-DTL-25677/1 adapter shall be provided on the tank to allow the tank to be filled by another tank and pump unit which is equipped with mating nozzle.

d. A visual means of displaying in one-half (1/2) pint increments, the amount of fluid pumped from the supply tank shall be provided. The volume of lubricant within the overflow tank shall be visually displayed in 1/2 pint increments and the display shall have an adjustable pointer (marker). If a sight glass is used, it must be guarded to prevent breakage during operation and storage

e. Dry weight of the unit shall be not greater than 96 pounds

f. A handle shall be provided on the overflow tank to enable the full tank to be

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conveniently carried and easily emptied.

3.18 Finish.

a. All painted surfaces shall be oil and corrosion resistant under all environmental conditions specified within.

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b. The color shall conform to FED-STD-595, Color No. 24052

c. Finishes of internal surfaces shall be corrosion resistant and compatible with the contractor selected materials and lubricants per 3.3.

3.19 Nameplate. The unit shall have a nameplate marked with the National Stock Number, Type Designator, and manufacturer's name and part number.

3.20 Instructions.. The unit shall have an attached operational instruction plate.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First Article Inspection. First article inspection shall be performed on one complete unit (see 3.1) This inspection shall include the examination and the tests specified in 4.5 through 4.5.7.5. (See Requirement/Verification Cross Reference Matrix, Appendix A).

4.3 Test Conditions. Unless otherwise specified, tests shall be conducted at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 70ºF). The MIL-DTL-25677/3 delivery nozzle shall be connected to a MIL-DTL-25677/1 adapter during all flow tests.

4.3.1 Test fluid. The test fluid shall be lubricating oil conforming to MIL-PRF-7808.

4.4 First article testing. One tank and pump unit shall be subjected to the tests specified in 4.5 through 4.5.7.5.

4.5 Test methods.

4.5.1 Examination of product. The first article unit shall be examined to confirm adherence to specified requirements such as configuration, design details, use of specified parts and marking. This examination shall consist of visual inspection of the unit, simple measurement of unit dimensions, weighing of the unit and review of the manufacturing drawings as required. Noncompliance with any specified requirements or presence of defects preventing or lessening

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maximum efficiency shall constitute cause for rejection.

4.5.2 Pressure test. The supply tank (including fittings) shall be subjected to an internal air pressure of 10 psi or 200 percent of the operating pressure, whichever is greater, for 10 minutes and the overflow tank (including fittings) shall be subjected to an internal pressure of 10 psi for a duration of 10 minutes. Any evidence of leakage shall be cause for rejection. The portion of the tank and pump unit on the discharge side of the pump, to and including the dispensing hose nozzle, shall be subjected to an internal hydrostatic pressure of 500 psi or that pressure produced by a force of 150 pounds applied at the pump handle, whichever is greater, for 5 minutes and at 20 psi for 5 minutes. Any evidence of leakage shall be cause for rejection.

4.5.3 Operational test. The supply tank shall be filled with lubricating oil conforming to MIL-PRF-7808. With the oil and the unit at a temperature of 70º ±5º F, the tank and pump unit shall discharge oil from the end of the fill nozzle at the rate of not less than 1 gpm with not more than 50 strokes/revolutions of the pump handle. The flow rate shall be based upon the time to pump 1.5 gallons of oil. At the start, during and after the test, all lines and hoses shall be full of oil and the relieving of pressure in the discharge line as specified in 3.4.e shall be demonstrated. The test shall be repeated four times with the last two tests being conducted at 2 ±0.1 gpm and 0.25 –0.0, +0.05 gpm respectively. The performance specified in 3.11 shall be demonstrated.:

4.5.4 Leakage test. The tank and pump unit with oil in the supply and overflow tanks shall be turned upside down before and after the unit has been built up to the operating pressure. There shall be no evidence of leakage.

4.5.5 Calibrated indicator tests.

4.5.5.1 Overflow tank test. The overflow tank shall be filled with measured quantities of 1, 3, 5, and 7 quarts. The indicator shall show the amount added to the 1, 3, 5, and 7 quart levels to within ±0.125 pints.

4.5.5.2 Supply tank test. The supply tank shall be filled to the 3-gallon level as shown on the indicator. The tank and pump unit shall then discharge oil from the end of the fill nozzle into graduated containers. The indicator shall show the amount of oil pumped from the supply tank at the 2, 3, 5, 7, 9, and 11 quart intervals to within ±0.125 pint.

4.5.6 Mobility tests. The unit (see 6.3.2) shall be towed by a vehicle having a pintle height of 10 to 30 inches at an average speed of 5 mph (maximum speed of 7.50 mph) for a distance of 5 miles over a course including right-angle obstructions 0.75 inch to 1.0 inches in height by 1.0 inch to 2.0 inches in width alternately spaced at 5 and 6-foot intervals at 60º angles to the direction of travel. At the completion of this test, the unit shall be inspected and there shall be no evidence of structural damage or loosening of any part of the unit. The unit, with the supply tank full of oil, shall then be dropped from a height of 1 foot onto a concrete floor until each of the resting points has impacted three times. The unit shall be inspected after each impact and there shall be no evidence of structural damage or loosening of any part. At the completion of these tests, the unit shall be able to dispense 3 gallons of oil in accordance with 3.11 at the 1 gpm at 50 cycles per minute rate.

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4.5.7 Environmental tests.

4.5.7.1 High temperature test. The unit shall be exposed to not less than 160ºF for not less than 48 hours. After that exposure, the temperature shall be reduced to not less than 125ºF. After the oil temperature in the supply tank has stabilized, the unit shall be removed from the environmental chamber and immediately the unit shall be able to dispense 3 gallons of oil at the 1 gpm at 50 cycles/minute rate.

4.5.7.2 Low temperature test. The unit shall be exposed to not more than –80ºF for not less than 48 hours. After that exposure, the temperature shall be raised to –40ºF. After the oil temperature in the supply tank has stabilized, the unit shall be removed from the environmental chamber and immediately the unit shall be able to develop not less than 300 psi deadhead oil pressure with not more than 50 pounds exerted on the pump handle then immediately the unit shall be able to deliver not less than 50 gpm with not more than 50 cycles/minute of the pump handle and with not more than 50 pounds exerted on the pump handle.

4.5.7.3 Humidity tests. The unit shall be subjected to 10 test cycles of humidity and temperature variations. Each cycle shall consist of 24 hours. Temperature and relative humidity fluctuations between 88ºF and 100% relative humidity to 105ºF at a relative humidity of 60% shall be conducted. A minimum of 3 hours shall be maintained at 105ºF and 60% relative humidity. The cycle shall be repeated ten times. Each test cycle can be terminated prematurely if deterioration is obvious after a visual examination of the unit. The unit's performance as specified in 3.11 shall not be affected by the humidity test.

4.5.7.4 Rainfall. The unit shall be subjected to rainfall rate equivalent to 4 in/hr at a wind velocity of 40 mph as commonly occur in areas of heavy rainfall. The droplet size shall not be smaller than 0.5 mm in diameter or larger than 4.5 mm. mph. All surfaces onto which rain can fall must be exposed to the test conditions. The unit shall be exposed for 6 hours and at 73°F (23C) and be capable of successful operation after the rain exposure. The unit's performance as specified by para. 3.11 will not be affected by the rainfall test.

4.5.7.5 Sand and dust particle tests. The unit shall be subjected to sand and dust concentrations that are at least 95% by weight silicon dioxide. The sand shall be of subangular structure with a mean Krumbein number (roundness factor) equal to 0.2 and a hardness factor of 7 mohs. The dust concentration for the blowing dust test shall be maintained at 10.6 + 7 g/m3. The configuration of the test item must reproduce, as closely as possible, the configuration that it would assume during storage or use, such as in a shipping container during deployment. The test shall be conducted for 90 minutes per face for the blowing sand test and 6 hours at 73°F (23C). The unit's performance as specified in 3.11 shall not be affected by the sand and dust tests.

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5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or ordering data (see 6.2i). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The PMU-29/E tank and pump unit is intended for dispensing oil to all aircraft engine and gear boxes requiring pressure-type oil servicing with synthetic base lubricating oil. The unit also dispenses corrosion preventive oil.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Packaging requirements (see 5.1).

6.3 Definitions.

6.3.1 Mission time. The mission time is 54 pump cycles and is determined by the time required for servicing a two-engine aircraft.

6.3.2 Unit. The unit is defined as the dolly with supply and overflow tanks, components, and towing handle.

6.3.3 Tank and pump assembly. The tank and pump assembly is defined as the supply tank, pump assembly, and related components.

6.4 Supersession data. This specification supersedes MIL-T-38266C(USAF) dated 10 July 1987.

6.5 Key Words.

Dolly mounted
Jet engine
Oil Services
Turbine engine

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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